RAG Assignment: Baseline vs. Re-Ranking

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Code: `https://github.com/Suniljadaun/rag_rerank`

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Setup

Document	GDPR, Regulation (EU) 2016/679 (EUR-Lex), 88 pages
Chunking	180 words per chunk, 40-word overlap → 419 chunks
Bi-encoder	`sentence-transformers/all-MiniLM-L6-v2` (384-dim), FAISS `IndexFlatIP`
Cross-encoder	`cross-encoder/ms-marco-MiniLM-L6-v2`
LLM	`gemini-2.5-flash`, temperature 0, identical prompt for both pipelines
Retrieval	Top 10 from the bi-encoder; top 3 passed to the LLM
Hardware	CPU (local laptop)

The GDPR has two parts that matter here: *recitals* (pages 1–31), which explain the reasons behind the law using "should", and *Articles* (from page 32), which are the binding rules using "shall". Both discuss the same topics with similar wording, which makes recitals natural distractors.

1 Architecture Comparison

A **bi-encoder** encodes the query and each chunk *separately*, turning each into one fixed 384-number vector. Relevance is the cosine similarity between the two vectors (an inner product, since the vectors are normalised). Because a chunk's vector does not depend on the question, all 419 chunk vectors are computed once in advance and stored in FAISS. At question time only the short query is encoded, so retrieval is very fast. The cost is that a whole chunk is squeezed into a single vector built without knowing the question, so fine distinctions, such as negation, exceptions, or whether a passage is a binding rule or an explanation, are blurred. Ranking is driven mainly by topical and word overlap.

A **cross-encoder** reads the query and one chunk *together* as a single input (`[CLS] query [SEP] chunk [SEP]`). In every transformer layer, each query token can attend to each chunk token (joint self-attention), and the model outputs one relevance score for the pair. This lets it notice that "when can a company *refuse* to delete" asks for exceptions, not for the grounds for deletion. Nothing can be pre-computed, because the score depends on the specific question, so it needs one forward pass per (query, chunk) pair.

The two are complementary. The bi-encoder gives **recall**: it cheaply narrows 419 chunks down to 10 likely candidates. The cross-encoder gives **precision**: it spends the expensive model only on those 10 and picks the best 3.

2 Qualitative & Rank Shift Analysis

For each query I found the *gold chunk*, the chunk that actually contains the answer, by keyword search and reading, before looking at the pipelines' output.

Category	Question	Gold	Baseline top 3	Re-ranked top 3	Gold rank
Keyword trap / recital distractor	How quickly must a company report a hack?	266	95, 94, 96	94, 267, 266	#6 → #3
Negation & exceptions	When can a company refuse to delete my data?	69	233, 234, 68	69, 233, 38	#5 → #1
Counter-example	When is a company not allowed to decide about me using only an algorithm?	243	76, 243, 78	76, 317, 79	#2 → #4
Recall limit	Can a company process someone's health data?	208	66, 44, 169	44, 66, 101	not in top 10

2.1 Case 1: Keyword trap / recital distractor (re-ranking helped)

The word "hack" never appears in the GDPR; the legal term is "personal data breach". The bi-encoder matched the general topic and ranked three recital chunks (95, 94, 96; recitals 85–87 on page 17) above the binding rule. Article 33 sat at rank #6 (chunk 266) and #5 (chunk 267). The cross-encoder moved chunk 267 from #5 to #2 and chunk 266 from #6 to #3.

Answer 1 vs. Answer 2. Both state the core rule: notify the supervisory authority without undue delay and within 72 hours, because recital 85 (chunk 94) repeats it. But Answer 1 is built only on non-binding recitals and **omits an essential clause**: the processor's duty to notify the controller without undue delay (Article 33(2), chunk 267). Answer 2 includes it and cites the Article. This is a degradation rather than a total failure: Answer 1 is not wrong, but it is incomplete and rests on explanatory text instead of the law.

2.2 Case 2: Negation & exceptions (re-ranking helped)

The question asks when deletion can be *refused*. The baseline top 3 was chunk 233 and chunk 234 (Article 17(1), the grounds on which data *must* be erased) and chunk 68 (start of recital 65, the right to be forgotten). These share almost every word with the query but answer the opposite question, which is a typical bi-encoder failure on negation. The chunk with the exceptions, chunk 69 (continuation of recital 65: "the further retention ... should be lawful where it is necessary ..."), was at #5. The cross-encoder moved it to #1.

Answer 1 vs. Answer 2. With only the erasure grounds, Answer 1 **misstated the rule**: it built "refusal" conditions by reversing two grounds from chunk 234 (consent withdrawn but another legal ground exists; the person objects but overriding legitimate grounds exist), and missed every real exception. Answer 2 listed all six exceptions from chunk 69: freedom of expression and information, legal obligation, public interest or official authority, public health, archiving or research or statistics, and legal claims.

One limitation: the binding version of these exceptions is Article 17(3) (chunk 235), which only moved from #9 to #7 and stayed outside the top 3 in both pipelines. Re-ranking fixed the answer's content, but via a recital.

2.3 Case 3: Counter-example (re-ranking hurt)

Here the baseline was already good: Article 22 (chunk 243) was at #2. Answer 1 was complete: the right not to be subject to solely automated decisions, all three exceptions in Article 22(2) (needed for a contract, authorised by law, explicit consent), and the safeguard of human intervention.

The cross-encoder pushed chunk 243 down to #4 and promoted chunk 317 (#5 → #2), a part of Article 47 on binding corporate rules that merely *lists* "the right not to be subject to decisions based solely on automated processing, including profiling in accordance with Article 22". That sentence repeats the query's idea almost word for word, but contains no rule about when such decisions are allowed. **Answer 2 lost two of the three exceptions** (contract and explicit consent) and

the human-intervention safeguard; it kept only the "authorised by law" exception from recital 71 (chunk 76).

Lesson: the cross-encoder can fall into its own keyword trap. It is a small 6-layer model trained on web search queries (MS MARCO), not on legal text, so re-ranking is not a guaranteed improvement and has to be evaluated per query.

2.4 Case 4: Recall limit (re-ranking cannot help)

Article 9 (chunk 208) starts with a ban: processing of health data "shall be prohibited", followed by exceptions such as explicit consent. It was not in the bi-encoder's top 10 at all, so the cross-encoder never saw it. Two likely reasons: the query says "health data" while Article 9 says "data concerning health", and chunk 208 begins with the end of Article 8 (child consent), which dilutes its vector. The retrieved chunks were recitals about medical-record access (chunk 66), public-interest health processing (chunk 44) and research (chunk 169).

Both answers misstated the rule: both said "yes, under certain conditions" and neither mentioned that the default is a prohibition. Answer 2 even inferred that a DPIA passage "could include companies", which is speculation. Possible fixes are a larger candidate pool (e.g. top 50 before re-ranking), hybrid keyword (BM25) plus dense search, chunking by Article instead of by word count, or a stronger embedding model.

3 Latency Profiling

Retrieval stages were timed over 8 questions × 5 repeats after one warm-up run, on CPU. LLM times are medians over the four case studies, counting successful calls only.

Stage	Median (ms)	p95 (ms)	Mean (ms)
Bi-encoder retrieval (embed query + FAISS top 10)	12.2	13.6	12.0
Cross-encoder re-rank (10 pairs)	307.0	331.3	308.5
Pipeline 1 retrieval total	12.2	13.6	12.0
Pipeline 2 retrieval total (retrieve + re-rank)	318.8	344.1	320.6
LLM generation, Answer 1	5234	–	–
LLM generation, Answer 2	5207	–	–

- Re-ranking 10 pairs is about **25 times slower** than the whole bi-encoder retrieval (307 ms vs. 12.2 ms), about 31 ms per pair.
- End to end, it matters much less: Pipeline 1 takes about 5.25 s and Pipeline 2 about 5.53 s, roughly **5% slower**, because the LLM call dominates.
- For Cases 1 and 2, that extra 0.3 s turned an incomplete or misstated answer into a correct one, which is a good trade for a legal question-answering tool. On a GPU the re-ranking cost would be smaller still.
- LLM time depends on network and server load, so it varies more between runs than the local retrieval stages.

4 Scaling Question

Measured: cross-encoding all 419 chunks for one query took 12,502 ms, i.e. 29.84 ms per (query, chunk) pair on CPU.

Collection size (chunks)	Extrapolated cross-encoder time per query
10,000	298 s (≈ 5 min)
1,000,000	29,838 s (≈ 8.3 h)
100,000,000	2,983,755 s (≈ 829 h)

We do not run the cross-encoder over the whole collection because its cost cannot be moved offline. Its score depends on the specific question, so every new query needs one full transformer pass per chunk, and the cost grows linearly with collection size. Already at 10,000 chunks a single query would take about 5 minutes, and at a million chunks about 8 hours.

The bi-encoder avoids this. Chunk vectors are computed once, offline. At query time we encode only the question and compare vectors, which is a cheap dot product; approximate indexes such as HNSW make even millions of vectors searchable in milliseconds. The cross-encoder then runs on a *fixed* 10 candidates, so its cost stays at about 0.3 s whatever the collection size.

The trade-off is that stage 1 sets a ceiling on recall: the re-ranker can only reorder what the bi-encoder found. Case 4 shows this directly. A larger k raises the ceiling but costs about 30 ms per extra candidate on this CPU (for example, $k = 50$ would take about 1.5 s), so k balances recall against latency.